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Bistability, Bifurcations, and Waddington's Epigenetic Landscape

Review

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Waddington's epigenetic landscape is probably the most famous and most powerful metaphor in developmental biology. Cells, represented by balls, roll downhill through a landscape of bifurcating valleys. Each new valley represents a possible cell fate and the ridges between the valleys maintain the cell fate once it has been chosen. Here I examine models of two important developmental processes — cell-fate induction and lateral inhibition — and ask whether the landscapes for these models at least qualitatively resemble Waddington's picture. For cell-fate induction, the answer is no. The commitment of a cell to a new fate corresponds to the disappearance of a valley from the landscape, not the splitting of one valley into two, and it occurs through a type of bifurcation — a saddle-node bifurcation — that possesses an intrinsic irreversibility that is missing from Waddington's picture. Lateral inhibition, a symmetrical cell–cell competition process, corresponds better to Waddington's picture, with one valley reversibly splitting into two through a pitchfork bifurcation. I propose an alternative epigenetic landscape that has numerous valleys and ridges right from the start, with the process of cell-fate commitment corresponding to the irreversible disappearance of some of these valleys and ridges, via cell-fate induction, complemented by the creation of new valleys and ridges through processes like cell–cell competition.

Introduction

In 1957, Conrad Hal Waddington published his famous drawing of the epigenetic landscape, which depicts how a cell progresses from an undifferentiated state to one of a number of discrete, distinct, differentiated cell fates during development [1]. The cell is represented by a ball, and it starts out in a valley at the back of the landscape (Figure 1). As the ball rolls forward and downward, the valley splits or bifurcates into two new valleys separated by a ridge. These new valleys represent alternative cell fates. External stimuli, such as inductive influences, or internal influences, such as homeotic genes, determine which of the two valleys a particular cell chooses. The valleys continue to split, and eventually the cell ends up in one of many terminal sub-valleys, which represent terminally differentiated states. The cell is held permanently in its terminally differentiated state by high valley walls. The steeper the walls, the more 'canalized', in Waddington's terminology, the cell fate.

Waddington's landscape was inspired by dynamical systems theory, and indeed it is more than just a metaphor. Each dimension of the landscape corresponds to a physical quantity. The left–right axis represents phenotype as depicted by a single time-dependent state variable (here termed x). This variable corresponds to a 'representative' output of the regulatory system that drives differentiation — for example, the concentration of a critical transcription factor. Of course a complete depiction of a cell's state would require many state variables, not just one, resulting in a hypersurface in a high-dimensional space. But hypersurfaces and high-dimensional spaces are difficult to visualize, and so Waddington restricted himself to one variable, with the hope that this oversimplification would nevertheless be a useful starting point for conceptually dissecting development [1].

The back-to-front y-axis represents time. The ball starts out at the back of the landscape and then moves continuously toward the front. One could also consider this dimension to represent a second state variable that drives development — an input to the regulatory system. If the y-axis is time, a ball will always eventually make it to the very front of the landscape, whereas if it is an input variable the ball may stop when this variable reaches its final value. For the purposes of this review, either view of this dimension will work fine.

The vertical z-axis represents the system's potential (Φ), a quantity analogous to an electrical potential or a gravitational potential (see Box 1 for a glossary of terms). The potential is a function of both x and y, and the shape of the potential surface is determined by the system of genes, RNAs, proteins, and metabolites that collectively control cell fate. In turn, the shape of the surface determines the dynamics of the cell's cell-fate regulators. A cell's journey from the back to the front of the landscape could be either completely deterministic, or deterministic except at the points where one valley splits into two, or stochastic throughout [2].

The view of development depicted in Waddington's epigenetic landscape is attractive and enduring. But, given what we now know about the logic of development and about dynamical systems theory, is Waddington's landscape a plausible picture of what really happens, at least in a qualitative sense?

Here I address this question by examining mathematical models of two basic developmental processes: cell-fate induction and lateral inhibition. I have kept the models simple, focusing on the essential features of the two processes and have tried to make the discussion self-contained, accessible to biologists without a background in dynamical systems theory and to dynamical systems theorists without a background in biology. Additional background information on the dynamical systems theory used here can be found in chapter 2 of Strogatz's textbook [3]. Further information on cell-fate induction and cell–cell competition can be found in chapter 3 of Gilbert's textbook [4]. For a historical perspective of these issues, the reader is referred to the pioneering work of Delbrück [5], Waddington [1], Jacob and Monod [6], and Kauffman [7]. Recent theoretical work relevant to potential landscapes in general and Waddington's landscape in particular can be found in [8–10], and additional modern perspectives on the topic can be found in [11–15].

The Connection between Differentiation and Multistability

At the start of a cell's journey through Waddington's landscape, the landscape consists of a single valley. This

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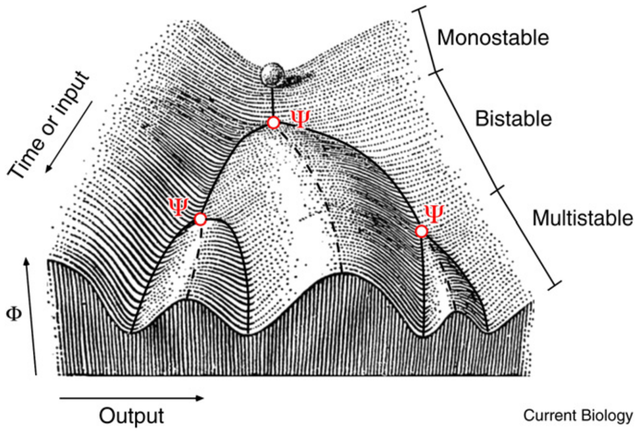

Figure 1. Waddington's epigenetic landscape (adapted from [1]).
Differentiating cells are represented by a ball rolling through a changing potential surface. At the outset of development, the ball can also represent a region of cytoplasm in a fertilized egg. New valleys represent alternative cell fates, and ridges keep cells from switching fates. At the back of the landscape, the potential surface is monostable — there is a single valley. When the valley splits the landscape becomes bistable and then multistable. The solid black lines represent stable steady states, the dashed lines represent unstable steady states, and the red circles represent pitchfork bifurcations (Ψ).

means that for a given slice of time, there is a single steady-state value for *x*, the *x* coordinate of the bottom of the valley. As time goes on, the landscape acquires additional valleys, always separated from each other by ridges. Thus, the system goes from being monostable — one valley — to bistable and then multistable (Figure 1). The idea that alternative cell fates may correspond to alternative stable steady states predates Waddington [5] and continues to be an important guiding principle for understanding differentiation. For a system to be bistable or multistable, it must include positive or double-negative feedback loops [16–18].

The relevance of bistability or multistability to biology received experimental support from studies of β-galactosidase induction in bacteria [19] and of the lysis–lysogeny decision in bacteriophage λ [20]. More recent experimental work has established bistability as the basis of the all-or-none, irreversible maturation of *Xenopus* oocytes [21–23]. Bistability appears to be at the heart of decisive, irreversible biological phenomena beyond cell differentiation as well, such as the transitions between phases of the cell division cycle [24–28].

Creating and Eliminating Valleys through Bifurcations

Waddington's landscape not only features multistability, but also specifies how valleys are created at the time of critical developmental transitions. The process of producing a new valley is termed a bifurcation, and the type of bifurcation seen in Waddington's landscape, where one valley turns into two valleys plus an intervening ridge, is called a pitchfork bifurcation (Figure 1). Note that the complete name for Waddington's bifurcation is a supercritical pitchfork; a pitchfork can also have a ridge splitting into two ridges plus a valley, which is termed a subcritical pitchfork. Here we will only encounter supercritical pitchforks, and so for economy's sake the term 'supercritical' is omitted.

A pitchfork bifurcation is not the only way of creating or eliminating valleys from a landscape. A new valley can arise somewhere far from the existing valleys, and an old valley can cease to exist by dead-ending rather than merging with another valley. These processes are termed saddle-node bifurcations, and although there are no saddle-node bifurcations on Waddington's landscape, we will encounter them soon.

In summary, Waddington's epigenetic landscape begins with a monostable system — one valley, corresponding to one possible stable steady state for the undifferentiated cell. As time goes on, the landscape goes through a succession of pitchfork bifurcations, giving rise to new valleys and new possible cell fates. With this in mind I formulate models of developmental processes, generate potential surfaces, and see how the surfaces compare with Waddington's picture — something that Waddington's classic 1957 book stopped short of doing. I begin with a model of cell-fate induction.

Cell-Fate Induction

In cell-fate induction, a cell or a group of cells produces an inductive stimulus that causes another cell to adopt a new phenotype. A schematic view is shown in Figure 2A. Well-studied examples include mesoderm induction in the early *Xenopus laevis* embryo [29], progesterone-induced maturation in *Xenopus* oocytes [30], R7 photoreceptor induction in the *Drosophila melanogaster* eye [31], and vulval induction in *Caenorhabditis elegans* larvae [32].

One key feature of cell-fate induction is that the inductive stimulus need not be maintained indefinitely; after some commitment point, the stimulus may be withdrawn and the cell will continue with its induced developmental program. Another is that induction results in an all-or-none switch between qualitatively distinct cell fates. Intermediate fates are seen only transiently. Both of these features suggest that positive feedback and multistability are involved in the process.

Modeling Cell-Fate Induction

To keep things simple and consistent with Waddington's use of a single variable for phenotype, I start with a single-variable model of cell-fate induction where some differentiation regulator, denoted *x*, promotes its own synthesis via a positive feedback loop (Figure 2B). The model is similar to one proposed for progesterone-induced *Xenopus* oocyte maturation [22,23,33], but for present purposes can be considered a generic model of cell-fate induction.

I assume that there is some basal rate of *x* synthesis, denoted by α_0, plus a feedback-dependent component of *x* synthesis. Because nonlinearity is important for the generation of bistability [34,35], and biological response functions are often well-approximated by Hill functions, I assume that the feedback-dependent rate of *x* synthesis is proportional to a Hill function with a high Hill coefficient (n = 5). Taken together:

$$\text{Synthesis rate} = \alpha_0 + \alpha_1 \frac{x^5}{K^5 + x^5} \qquad \text{[Equation 1]}$$

where α_1 is the maximum rate of feedback-dependent synthesis of *x* and *K* is the concentration of *x* where the

Box 1

Glossary of terms.

Attractor: Stable steady states are attractors. In multivariable systems there are other types of attractors as well, including stable limit cycles and strange attractors.
Bifurcation: A splitting of one thing into two. In dynamical systems theory the term refers to the splitting of steady states or fixed points. For one-variable systems, there are three classes of bifurcations: saddle-node bifurcations, pitchfork bifurcations, and transcritical bifurcations. All of these bifurcations arise in systems with positive feedback. Examples of saddle-node and pitchfork bifurcations are provided in the text. An example of a transcritical bifurcation is provided by a system with Michaelian, rather than sigmodal, positive feedback;
e.g. $dx/dt = \alpha(x/1+x) - x$. As α increases above 1, the system goes from having a single stable steady state at $x = 0$, to two steady states, an unstable one at $x = 0$ and a stable one at $x = \alpha - 1$. The result is a linear response with a threshold.
Bistability: Having two stable steady states or two potential wells.
Epigenetics: Waddington coined the term and defined it as the science concerning how different discrete cell fates arise from a single set of genetic instructions. A cell's epigenetic state is generally stable, heritable, and associated with specific patterns of histone and DNA 'marks' or post-translational modifications.
Hill function: A function of the form:

$$y = \frac{x^n}{K^n + x^n}$$

For $n > 1$ the Hill function yields a sigmoidal curve. Ultrasensitive and cooperative regulatory processes often exhibit sigmoidal steady-state responses and, while the derived expressions for these responses are generally not Hill functions, the Hill function usually provides a simple, reasonably accurate approximation for these sigmoidal responses. The parameter K is the concentration of x at which the response is half-maximal. The exponent n determines how switch-like the response is.
Multistable: Having more than one stable steady state. In practice, the term 'bistability' is usually used if there are two such states, and 'multistability' is reserved for more than two.
Potential: For these purposes, I define potential by analogy. Consider a real landscape, with a ball starting out at some point on the landscape. The altitude of the ball is its potential, and the steepness of the slope — the spatial derivatives of the potential — gives the force on the ball. Thus, a vector field of forces can be calculated from a scalar field of potentials by taking derivatives. For biochemical reactions one begins with a field of velocities rather than a field of forces, and one can define the potential as a function which, when differentiated, yields this velocity field.

feedback is half-maximal. I assume that x is degraded through a simple mass action process:

$$\text{Degradation rate} = \beta x \qquad \text{[Equation 2]}$$

Combining these two expressions yields an ordinary differential equation for the net change of x with time:

$$\frac{dx}{dt} = \alpha_0 + \alpha_1 \frac{x^5}{K^5 + x^5} - \beta x \qquad \text{[Equation 3]}$$

For the right choice of parameters, x can be bistable. This is perhaps most easy to see graphically. x's rates of production (Figure 2C, red curve) and degradation (Figure 2C, blue curve) are plotted as a function of x. Steady states are found where the production and degradation rates are equal and the two curves intersect. It is easy to choose parameters such that the sigmoidal red curve snakes around the blue curve, intersecting it in three places. Two of the intersections correspond to stable steady states, one with $x = 0$ and the other with $x \approx 1.7$, and the middle one corresponds to an unstable steady state.

Calculating the Potential Surface

Next the steady states are depicted in a Waddington-like potential framework. The potential Φ is defined as a function whose derivative $d\Phi/dx$ yields the speed at which x moves toward its steady state. By anti-differentiating, it follows that:

$$\Phi = -\int \alpha_0 + \alpha_1 \frac{x^5}{K^5 + x^5} - \beta x \, dx \qquad \text{[Equation 4]}$$

The minus sign in front of the integral ensures that a positive slope will make x move in the negative direction, to the left. Evaluating the integral in Equation 4 yields an algebraic expression that is too complicated to spell out here, but that has a simple shape when plotted (Figure 2D). The two stable steady states sit at the bottoms of two valleys, with the left-hand stable state being the global minimum of the potential. The unstable steady state sits at the top of a ridge. Both valley bottoms represent attractors; balls that start out to the left of the ridge top are attracted to the bottom of the left-hand valley, and those to the right of the ridge top to the bottom of the right-hand valley. And finally, the steepness of the valley walls, or the canalization of the cell fate, in Waddington's terminology, is determined by the nonlinearity of the positive feedback: the higher the Hill coefficient, the steeper the valley walls, making the cell fate more robust with respect to perturbations in x.

Note that there are some differences between how a cell moves left or right on our landscape (or Waddington's) and how a real ball would roll or slide down a real mountainside. In the latter case, the slope of the mountain $(d\Phi/dx)$ determines the acceleration of the ball. Thus, a ball picks up speed when it rolls down a constant slope. In the case of biochemical reaction networks, however, $d\Phi/dx$ determines x's velocity rather than acceleration, so that a ball rolling down a constant slope would travel at constant speed. This is the way a ball would move if it slid down a mountainside through a highly viscous medium. When the ball reached the bottom of a potential well, it would not just stop accelerating, it would stop moving.

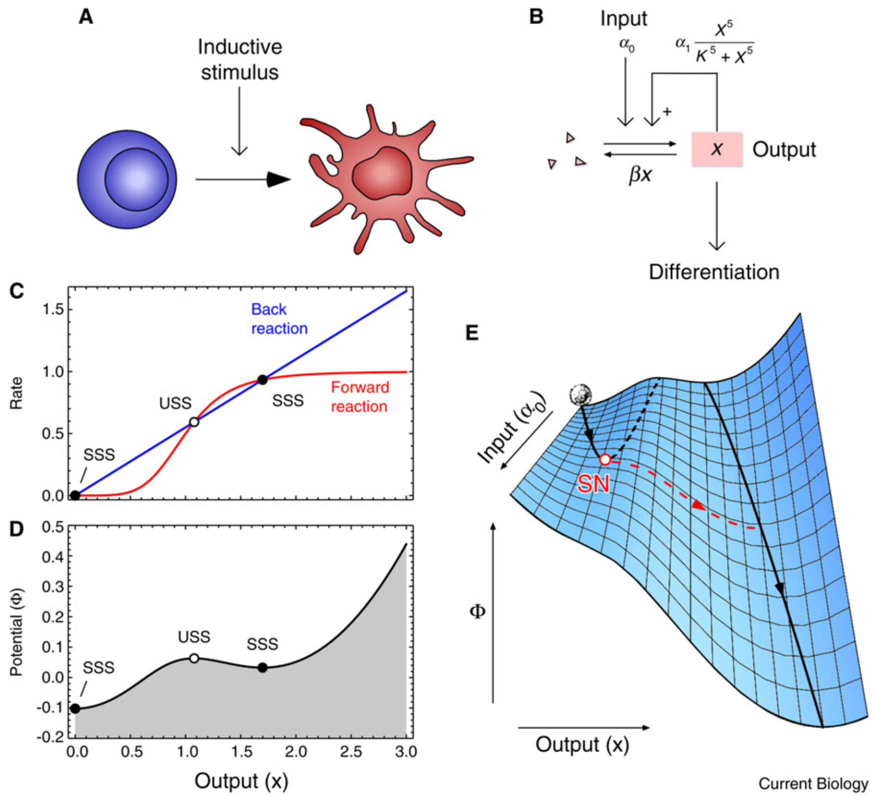

Figure 2. Cell-fate induction.

(A) Schematic view of a cell-fate induction process. (B) A simple mathematical model of a positive feedback system that could trigger and maintain differentiation, where some differentiation regulator, denoted *x*, promotes its own synthesis via a positive feedback loop. (C) Rate–balance analysis. The blue line depicts the rate of the back reaction as a function of *x*. The red curve depicts the rate of the forward reaction. The intersections of the two define where the forward and back reaction rates are balanced and the system is in a steady state. The filled circles are stable steady states (SSS) and the open circle is an unstable steady state (USS). (D) The potential function. The stable steady states lie at the bottoms of valleys and the unstable steady state sits at the top of a ridge. Values were chosen for the parameters in Equation 3 as follows. I initially assumed the input $\alpha_0 = 0$. I assumed $K = 1$, which is equivalent to choosing units for the *x*-axis so that 1 unit of *x* produces a half-maximal forward reaction rate when the input is 0. I assumed $\alpha_1 = 1$, which is equivalent to choosing units for the *y*-axis so that the maximal rate of the forward reaction when the input is 0 is 1. Finally, I assumed $\alpha_1 = 0.55$, which makes the curves interact three times and makes the system bistable. (E) A saddle-node epigenetic landscape. The input (α_0) increases from 0 to 0.9 as time increases, and the potential is calculated according to Equation 4. Solid lines denote stable steady states (valley bottoms), and the dashed line represents the unstable steady state (the top of the ridge between the two valleys). Cell-fate commitment occurs when the left-hand valley and the ridge meet each other and disappear at a saddle-node bifurcation (SN).

Cell-Fate Induction Eliminates Valleys Rather Than Creating New Valleys

If nothing changes about the equation that describes the synthesis and destruction of *x* (Equation 3), then nothing will change about the landscape; it will continue to have two valleys and a ridge, in perpetuity.

But suppose the cell becomes exposed to an inductive stimulus, an input to the regulatory system (Figure 2A). In terms of Equation 3, the input could be taken as an increase in the basal rate of *x* synthesis, α_0. As the input increases, the shape of the landscape changes. If, for the sake of simplicity, I assume that the input increases relatively slowly, the resulting two-dimensional potential landscape can be considered to be a succession of one-dimensional slices. The result is shown in Figure 2E. The landscape begins with two valleys, with the left-hand valley being deeper than the right-hand valley. As the stimulus begins to increase, the landscape tilts, with the right side becoming progressively lower with respect to the left. As a result, the bottom of the left-hand valley begins to shift to the right and the ridge top that separates the valleys begins to shift to the left. Eventually the left-hand valley and the ridge meet and disappear at a saddle-node bifurcation (Figure 2E, labeled SN). Once the left-hand valley is no longer a valley, a cell that starts out in the left-hand (uninduced) valley is forced to roll downhill, to the right, settling at the bottom of the right-hand valley. Thus, the cell leaves the uninduced cell fate and adopts the induced cell fate, because the valley corresponding to the uninduced cell fate no longer exists.

Why does the inductive stimulus α_0 tilt the landscape to the right (Figure 2E), making the left-hand valley disappear? Consider again the expression for the potential function Φ (Equation 4). The stimulus α_0 enters the integral only in the first term; integrating this term yields $-\alpha_0 x$. This is a flat line when $\alpha_0 = 0$, and as α_0 increases, it acquires an increasingly large negative slope. In this way the stimulus progressively tilts the potential surface, dumping the cell out of the left-hand potential well.

Although it was natural to assume that the induction stimulus acts by increasing the value of α_0, I could have alternatively made the stimulus act through any of the other parameters of Equation 4 — for example, by decreasing the value of the degradation constant β in the face of a small (but non-zero) constant value for α_0. Would this alter the conclusion that cell-fate commitment occurs as a result of the disappearance of a valley at a saddle-node bifurcation? The answer is no. No matter how I choose to have the inductive stimulus affect the model, the result is the same. The cell commits to the induced fate because the valley corresponding to the uninduced fate disappears through a saddle-node bifurcation.

Finally, I should note that just as valleys can be eliminated from a landscape through a saddle-node bifurcation, they can also be created through a reverse of the process depicted in Figure 2E. But this reverse process still does not resemble the type of bifurcation depicted in Waddington's landscape. The new valley does not split off from the initial valley — it is born somewhere else in state space. And a cell will not be able to move into the newly created valley until its own valley disappears in another saddle-node bifurcation.

Differences between the Saddle-Node Landscape and Waddington's Epigenetic Landscape

The differentiation process represented on the alternative epigenetic landscape (Figure 2E) is very different from that which takes place on Waddington's landscape (Figure 1). Here the induced cell fate is present in the landscape before the cell-fate commitment occurs; in Waddington's picture, the new cell fate splits off from the old one. Here the process of cell-fate commitment corresponds to the disappearance of a valley from the landscape; in Waddington's picture, it corresponds to the creation of a new valley. And finally, the bifurcations involved are different. Here the valley disappears through a saddle-node bifurcation, whereas in Waddington's picture the new valley is born through a pitchfork bifurcation. These differences have important implications for the basic character of the developmental transition.

The Instability of Intermediate States

In Waddington's model, immediately after a cell has committed to a fate, the valley in which it sits is very close to the valley not chosen (Figure 1). The steady-state values of x for the two alternative states are close together. If, at this point in the differentiation process, all of the inputs into the cell-fate decision are held constant, the cell should remain in this slightly differentiated state indefinitely — this intermediate state is stable. To get the chosen valley to diverge further from the valley not chosen, one would need to further increase the input that is driving the cell-fate decision. If one could artificially hold the input at various slightly different levels, the result would be cells arrested in slightly different differentiation states. A graded, continuously variable stimulus would give rise to a continuously variable response.

In contrast, intermediate differentiation states can be attained only transiently in the saddle-node landscape (Figure 2E). Before the saddle-node bifurcation, if you were to force a cell to have an intermediate level of x, it would move either right or left, back towards one or the other of the valley bottoms. After the bifurcation, if the cell were to start at an intermediate level of x it would move right towards the bottom of the single valley. But in neither case would it be possible for the cell to remain indefinitely in an intermediate state. Intermediate states are unstable.

The question then is whether in real cell-fate induction processes, intermediate states are stable or unstable. For many such processes the answer is difficult to determine experimentally because it would be difficult to hold the inductive stimulus at a constant, intermediate level. However, for *Xenopus* oocyte maturation, which can be studied *in vitro*, the answer is clear: constant, intermediate concentrations of the inductive stimulus progesterone cause oocytes to either mature fully or to not mature at all [23]. Intermediate phenotypes can be obtained only transiently; they thus appear to be unstable. This finding fits better with the saddle-node landscape model than with Waddington's model.

The Irreversibility of Cell-Fate Commitment

The saddle-node bifurcation provides the cell-fate commitment process with an intrinsic irreversibility that is missing from Waddington's landscape. Once the ball has rolled down into the right-hand valley (that is, the cell has committed to the induced cell fate), one can remove the inductive stimulus and the cell will tend to remain in the same valley (Figure 2E). The way in which the potential surface folds adds an arrow of time to the process of cell-fate commitment. This is not the case with a pitchfork bifurcation; remove the stimulus and the two valleys merge back together, allowing the differentiated cell to return to its undifferentiated state.

Again, *Xenopus* oocyte maturation provides an experimental test of these ideas. When progesterone is washed away from mature oocytes, they remain mature; they do not de-differentiate [22]. Once again this finding fits better with the saddle-node landscape model than with Waddington's model.

The Narrowing of Developmental Potential during Differentiation

As normal development proceeds, the developmental potential of a cell typically decreases from totipotency to pluripotency, multipotency, oligopotency, and unipotency. This narrowing of the developmental potential of a cell is particularly well worked out for the hematopoietic lineages [36]. For example, during differentiation an uncommitted hematopoeitic stem cell might first commit to one of two broad classes of ultimate fates (myeloid vs. lymphoid), then to one of a handful of fates (e.g. erythrocytic/megakaryocytic rather than granulocytic), and then finally to a specific terminally differentiated fate. Both Waddington's epigenetic landscape and the saddle-node landscape account for this phenomenon, but in different ways. On the Waddington landscape, a differentiating cell loses developmental potential because, once it passes a pitchfork bifurcation, ridges prevent it from shifting to a different fate. Alternative fates remain present somewhere on the landscape, but the cell cannot get to them. In contrast, on the saddle-node landscape some potential fates disappear completely; the cell's developmental potential is narrowed because alternative fates are gone.

These two models can, in principle, be tested by artificially manipulating a cell to place it somewhere else on the epigenetic landscape, then let it go and see where it ends up. For example, one could inject an undifferentiated cell with a pulse of protein x, moving the cell somewhere to the right of its normal starting point. In the case of Waddington's landscape, the ball should roll back down to the bottom of the single valley; only the uninduced cell fate is available to the cell at this point in development. In the case of the alternative landscape described here, if the move puts the cell beyond the ridge top, it will roll down into the next valley and adopt the alternative, induced cell fate.

Again, this type of experimental manipulation has been carried out in studies of *Xenopus* oocyte maturation. The key state variable x can be taken to be the protein kinase Mos, which is present at unmeasurably low concentrations in the immature oocyte. The stimulus is the hormone progesterone, which increases the rate of Mos translation and causes the oocyte to mature into a stable, high-Mos state. The question then is what happens if an oocyte is not treated with progesterone, but instead is injected with Mos protein or mRNA? Will it move back to the uninduced state once the injected protein/mRNA is gone, or will it stay in the high-Mos, induced state? The answer is the latter [37], consistent with the idea that both a low Mos valley and a high Mos valley were present on the oocyte's epigenetic landscape before progesterone was applied.

The converse experiment would be to take a cell that has committed to an induced cell fate — it has progressed past the pitchfork bifurcation in the Waddington landscape or the saddle-node bifurcation in the alternative landscape — and, without removing the inductive stimulus, transiently decrease the concentration of x, which would move the cell to the left on the landscape. In the Waddington picture, there are two valleys past the bifurcation point, so the cell would end up in the left-hand valley, which corresponds to a different cell fate. In the saddle-node landscape, the left-hand valley has disappeared, so that the cell has no choice but to move back to the bottom of the right-hand valley. To our knowledge this experiment has not been carried out; certainly it would be an informative test of Waddington's landscape vs. the saddle-node landscape.

Manipulating the concentration of x in an undifferentiated or differentiated cell can be viewed as cell-fate reprogramming. Waddington's epigenetic landscape suggests that reprogramming a differentiated cell should produce more fates than reprogramming a less-differentiated cell would. The saddle-node landscape suggests the opposite. The art of reprogramming cells has been advancing rapidly; it will be interesting to see which prediction wins out.

From One State Variable to Many

Note that real cell fate control networks contain multiple complicated feedback loops and dozens of relevant state variables [13,38]. Nevertheless, given the example of the one-variable model described above, and the experimental evidence supporting the view that differentiated cell fates coexist with the undifferentiated cell fate before cell-fate commitment occurs, I conjecture that cell-fate induction generally corresponds to the loss of one valley from the epigenetic landscape. I also conjecture that the valley is lost through a saddle-node bifurcation, which allows the induced phenotype to be qualitatively distinct from the uninduced phenotype (the valley that the cell moves to once the uninduced valley is eliminated need not be close to the uninduced valley) and makes the process of induction difficult to reverse. If a cell-fate induction process is stochastic rather than deterministic — that is, if noise allows a cell to make occasional excursions up over ridges — then that cell may switch to the induced cell fate before the saddle-node bifurcation occurs, but it will have the option of switching back to the uninduced fate by wandering back over the ridge until the bifurcation occurs.

A Pitchfork Bifurcation in a Model of Lateral Inhibition

Waddington's landscape features pitchfork bifurcations rather than saddle-node bifurcations. This raises the question of what sorts of physical phenomena involve pitchfork bifurcations, and whether these bifurcations might be relevant in some developmental processes.

In physics, the archetypal example of a pitchfork bifurcation is the bending of a slender wooden ruler [3,39]. Suppose the ruler is standing on one end and a force is applied to the top. If the force is small, the ruler will compress slightly and the middle of the ruler will remain directly under the top of the ruler. But if the force is increased sufficiently, the ruler will bow one way or the other. As the force continues to increase, the bowing (in whichever direction it began) increases.

The ruler has gone through a pitchfork bifurcation. At low forces, there is one stable steady state (straight up and down; one valley in the potential surface). Beyond the bifurcation, the straight configuration remains a steady state but becomes unstable (it becomes a ridge), and two symmetrical stable steady states (two valleys representing bowed left and bowed right) appear. The key to the pitchfork bifurcation is the symmetry of the ruler.

Might such a symmetrical situation be relevant in development? The answer is yes. One example, quite analogous to the bending ruler, is the deformation of a flat epithelium into one with thickenings and infoldings [39]. Another, which I will examine here, is lateral inhibition, a process where new fates are created through cell–cell competition. Lateral inhibition is important for producing patterns of alternating cell fates in a tissue, so that a cell of one type will be surrounded by cells of another type, and for stem cell reproduction where one daughter maintains an ability to reproduce while the other goes on to a differentiated fate. The best-studied example of lateral inhibition involves two cell-surface proteins, Delta and Notch, where Delta can be regarded as a ligand and Notch as a receptor. High levels of Delta on one cell repress, through the intermediacy of Notch, the expression of Delta in the cells that the first cell contacts.

Suppose there is a mother cell that expresses high levels of Delta and high levels of (unengaged) Notch. Then suppose the mother cell divides symmetrically into two identical daughter cells, each possessing half of the Delta molecules and half of the Notch molecules. What will happen to Delta expression in the competing daughter cells? Intuitively, it seems like one cell should win out and become a high Delta-expresser, like the mother cell was, and the other should end up in a low Delta state (Figure 3A).

To explore this further I examine a simple model of mutual inhibition, following the lead of Collier and colleagues [40]. In the mother cell the Delta protein is represented by x, and it is produced at some basal rate α and degraded by a first-order process with a rate constant of β:

$$\frac{dx}{dt} = \alpha - \beta x \qquad \text{[Equation 5]}$$

Once the daughters are born, the dimension of the system is doubled: x becomes two x variables, x_1 and x_2, representing the Delta molecules in each of the two cell. I assume that x_1 inhibits the production of x_2 in daughter 2, and, conversely, x_2 inhibits the production of x_1 in daughter 1 (Figure 3B). Using an inhibitory Hill function to describe this interaction, and assuming that the rate constants are the same in the two cells, leads to a model with two interlinked ordinary differential equations:

$$\frac{dx_1}{dt} = \alpha \frac{K^n}{K^n + (I \cdot x_2)^n} - \beta x_1 \qquad \text{[Equation 6]}$$

$$\frac{dx_2}{dt} = \alpha \frac{K^n}{K^n + (I \cdot x_1)^n} - \beta x_2 \qquad \text{[Equation 7]}$$

The parameter I represents the strength of the interaction between the two cells, or the fraction of the Delta molecules on cell 1 that are in contact with Notch molecules on cell 2 (and vice versa). This parameter can be considered to be the input that drives the process of competition and differentiation.

The steady states of the system can be solved for and plotted as a function of the input I (Figure 3C). When I is zero, the steady-state levels of x_1 and x_2 are both 2 surface

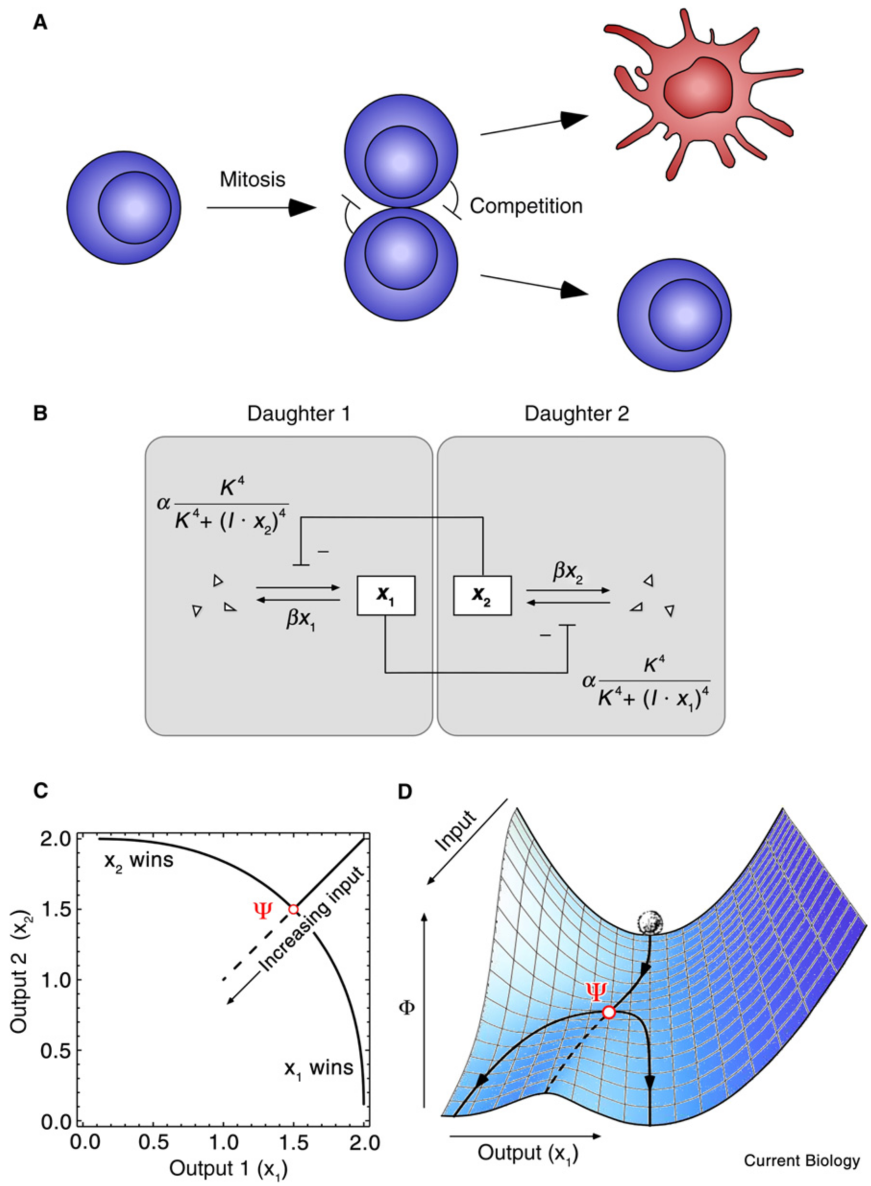

Figure 3. Lateral inhibition: competition between two daughter cells for expression of a signaling molecule.

(A) Schematic view, with the blue cells possessing high levels of Delta and the red cell possessing a low level of Delta. (B) Mathematical depiction of the mutual inhibition of the synthesis of a Delta-like protein x by x in a neighboring cell. (C) Steady states in the lateral inhibition model. The parameters chosen for the model (Equations 6 and 7) are $K = 1$, which defines the units for x_1 and x_2; $\alpha = 1$, which defines the units for the rates; and $n = 4$ and $\beta = 0.5$, arbitrary choices that make the system bistable. The interaction strength I is taken as the input to the system, and is varied from 0 to 1. At $I = 0.5$, the system goes through a pitchfork bifurcation (denoted Ψ). (D) The potential landscape of a lateral inhibition process. At the pitchfork bifurcation, one cell goes one way and the other one goes the other.

concentration units, the same as was present in the mother cell. As I increases, the levels of x_1 and x_2 decrease in a symmetrical fashion until I reaches a critical value of 0.5 units. At that point, the single stable steady state splits into two stable steady states, one with x_1 dominating and one with x_2 dominating, and an unstable steady state where the two x levels are equal (Figure 3C). This system has gone through a pitchfork bifurcation. As the interaction strength continues to increase, the concentration of x in one daughter cell rises back toward what it was in the mother cell — it returns to the undifferentiated state — and the concentration of x in the other daughter cell approaches zero, allowing it to differentiate.

The key here is that the system is symmetrical, and the input stimulus that drives the differentiation (I) impinges symmetrically on the two cells. Any imperfection in the symmetry will change the pitchfork bifurcation into a saddle-node bifurcation. If I had chosen different parameters for each cell or assumed that the stimulus impinges upon only one cell — say, by varying the value of one of the two α's — the result would have been a saddle-node bifurcation. Thus, the type of bifurcation depends both upon how the system is wired and how the stimulus affects the system.

The next step is to display this behavior in a landscape framework. It turns out that potential functions are not easy to calculate or interpret for two-dimensional systems where the variables are as interconnected as they are in this model [8]. A reasonable way forward is to display only one of the two output variables on the landscape (say, x_1), and calculate the potential function along a one-dimensional slice of the two-dimensional x_1 x_2 space. The result is shown in Figure 3D.

As the interaction strength increases and cells progress through the valley, the valley floor shifts to the left, towards lower values of x_1. At a critical value of the interaction strength I, the one valley splits into two valleys (solid lines) and a ridge (dashed line) through a pitchfork bifurcation (Ψ). This bifurcation represents a symmetry-breaking event where either daughter cell 1 or 2 begins to dominate the other in terms of its expression of the Delta-like protein x. Daughter cell 1 chooses one valley or the other and the value of x_1 either returns to the high value that was present in the mother cell, or progresses towards a low value of x_1. Daughter cell 2 chooses the other valley. Thus, the lateral inhibition model has yielded a potential landscape that closely resembles Waddington's epigenetic landscape.

It may be the case that the mother cell pre-patterns itself into a polarized cell that gives rise not to two identical daughters, but to a high x daughter destined to remain undifferentiated and a low x daughter destined to differentiate. In this case, the bifurcation occurs prior to cell division through an intracellular competition between regions, rather than an intercellular competition between daughters. But the basic logic of the process is the same, with the splitting of the cell into two regions corresponding to the Waddington-like pitchfork bifurcation.

Note that this pitchfork bifurcation is intrinsically reversible. If the interaction strength is returned to zero (for example, if the cells are separated from each other), both cells will return to their original high x state. To lock a low x cell into this differentiated state, the cell would need to proceed through a saddle-node bifurcation and fall into another valley that does not provide a route back to the undifferentiated state. Indeed, Wang and co-workers [10] added two extra autocatalytic positive feedback loops to a system shown in Figure 3B in order to produce a model of lateral inhibition where the cell-fate choice was hysteretic or irreversible. Moreover, there is experimental evidence showing that when *Drosophila* ovarian germ line cells divide to form one germ cell and one cystoblast — a process like that shown in Figure 3A — an intracellular positive feedback loop locks the cystoblast fate in place. The irreversibility of this cell-fate choice has been proposed to be due to traversal of a saddle-node bifurcation [41]. Thus, even in symmetrical processes like lateral inhibition, the need to make the induced fate irreversible may mean that cell fates are ultimately disappearing from the epigenetic landscape through saddle-node bifurcations.

Conclusions

Despite the complexity of development, I agree with Waddington that a simple epigenetic landscape picture is useful for understanding the essence of developmental transitions. And, as is implicit in Waddington's picture, I agree that alternative differentiated states are likely to correspond to multiple stable steady-states of the dynamical system that drives development. However, there are fundamental, qualitative differences between Waddington's view of what happens to the landscape during differentiation, and what happens to the landscape in the models analyzed here. The bifurcations are different, and the relationships between the bifurcations and biological processes like cell-fate commitment are different, and in my opinion these differences are important.

In Waddington's epigenetic landscape, a cell begins at the bottom of a single potential well, and then, as development proceeds, the one well successively splits into many more, representing the possible differentiation states of the cell. I propose two modifications to this view. First, I propose that differentiation mainly involves the disappearance of valleys from the landscape, not the appearance of new valleys. This view is consistent with simple mathematical models of cell-fate induction and with experimental studies of cell-fate induction processes. Second, I propose that the disappearance of the valleys occurs through saddle-node bifurcations, which provide an intrinsic irreversibility to the process of differentiation, a type of irreversibility missing from Waddington's original picture.

Finally, I show that the processes depicted on Waddington's original landscape correspond to intrinsically reversible pitchfork bifurcations, which could correspond to symmetry-breaking, intrinsically reversible developmental events like the generation of new cell fates through cell–cell competition. Even in these cases, however, the cell will not be locked into its new fate until a valley disappears from the landscape through a saddle-node bifurcation. This fact underscores the importance of the saddle-node bifurcation in development and differentiation.

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